Journal of Plant Growth Regulation (2026) 45:861–875
https://doi.org/10.1007/s00344-025-11925-y

Genetic and Agronomic Approaches to Micronutrient Biofortification in Solanaceae Plants

Aiana[1] · **Santosh Kumar Upadhyay**[2] · **Kashmir Singh**[1]

Received: 7 July 2025 / Accepted: 15 October 2025 / Published online: 27 November 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

Micronutrient deficiencies, particularly involving iron, zinc, and provitamin A, represent a pressing global health concern. Biofortification emerges as a sustainable approach to enhance the nutritional profile of staple crops through genetic advancements, molecular breeding, and agronomic practices. Solanaceae crops, such as potatoes and tomatoes, offer a promising avenue for biofortification due to their widespread consumption, high adaptability, and genetic versatility. This review delves into recent progress in biofortification methodologies, emphasizing genetic engineering, CRISPR/Cas genome editing, QTL mapping, and transgenic technologies to improve micronutrient content in Solanaceae crops. Furthermore, it evaluates the contributions of soil and foliar fertilization, microbial-mediated biofortification, and nanoparticle-driven nutrient delivery systems. Challenges related to commercialization and societal acceptance of biofortified Solanaceae crops are also discussed. By harmonizing modern breeding techniques with cutting-edge biofortification strategies, solanaceae crops hold significant potential to advance global nutrition security.

Keywords Potato · Tomato · Iron deficiency · Zinc deficiency · Iodine deficiency · Vitamin A deficiency

Introduction

Global malnutrition affects millions of people worldwide, posing a significant public health challenge, particularly in low- and middle-income countries. By 2030, it is projected that nearly 600 million people will suffer from chronic undernourishment (FAO 2023a). Every person has the inherent right to obtain a nutritious diet. In 2021, more than 3.1 billion people globally, representing 42 percent of the world's population, were unable to afford a healthy diet. In 2022, 148 million children under the age of five were stunted, 45 million were wasted, and 37 million were overweight, with the majority of these cases occurring in low-income and lower-middle-income countries (FAO 2023b) (Fig. 1). The issue of malnutrition is further intensified by the growing global population, which is expected to hit 8 billion by 2030, with the majority of this growth happening in developing countries.

The Sustainable Development Goals (SDGs), established by the United Nations in 2015, aim to eliminate all forms of hunger and malnutrition by 2030. The initiative seeks to ensure that everyone, especially children, has year-round access to sufficient and nutritious food. Although progress towards achieving these goals is lacking globally, it is evident that developing nations and the world's most impoverished and vulnerable populations are suffering the most from our collective shortcomings (UN 2023). Although progress has been made in combating micronutrient deficiencies through dietary enrichment and food fortification, new approaches are needed to enhance the reach of nutrition programs in impoverished rural communities and to aid in the sustainable reduction of micronutrient deficiencies in rapidly growing urban populations.

To tackle these issues, biofortification has emerged as a promising strategy. Biofortification involves creating staple crops that are rich in micronutrients through the use of both traditional breeding methods and contemporary biotechnological techniques (Nestel et al. 2006). Biofortification offers

Communicated by Pramod Kumar Nagar.

✉ Kashmir Singh
kashmirbio@pu.ac.in; kashmir123@gmail.com

[1] Department of Biotechnology, Panjab University, Sector 25, BMS Block I, Chandigarh 160014, India

[2] Department of Botany, Panjab University, Sector 14, Chandigarh 160014, India

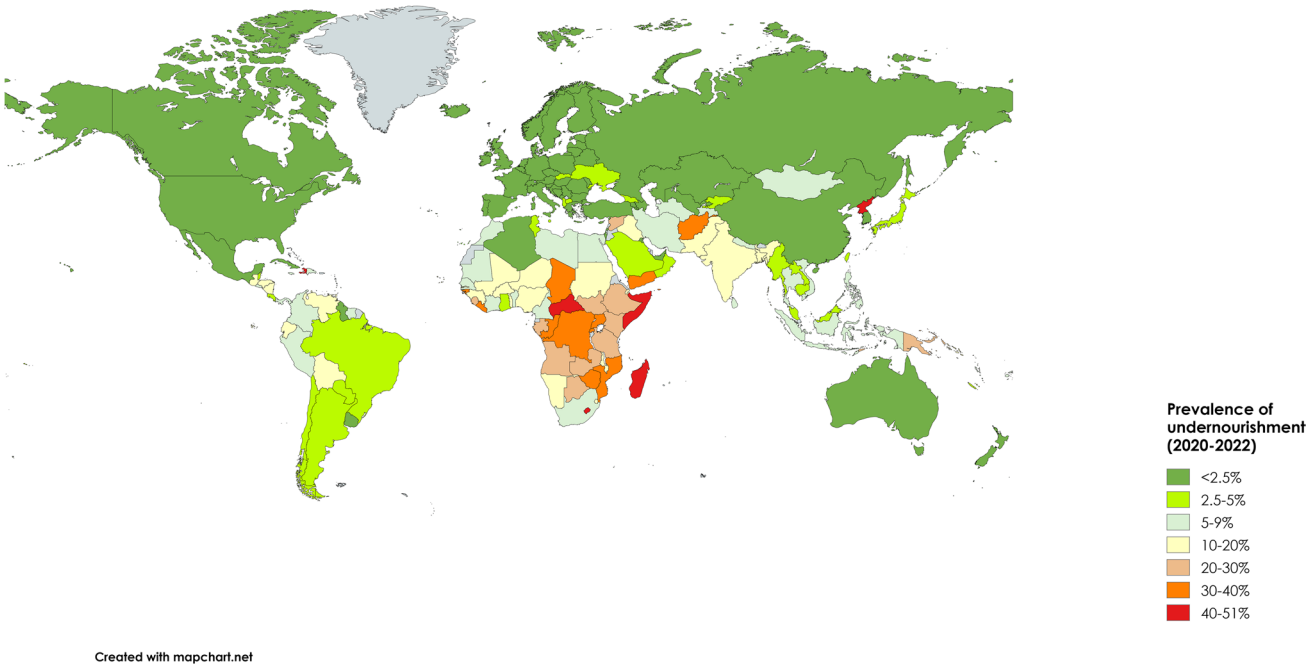


Fig 1 Prevalence of undernourishment (% of population) across the globe during 2020-2022 (data taken from FAO)

a practical approach to assist malnourished rural communities who might lack access to varied diets, nutritional supplements, and commercially fortified foods. Biofortification is regarded as a cost-efficient method to address micronutrient deficiencies among rural communities in developing nations, where this issue is most widespread (Mayer et al. 2008). At present, there are three widely used methods: agronomic biofortification (nutrient delivery through fertilizers and soil management), conventional biofortification (traditional plant breeding for enhanced micronutrient accumulation), and transgenic/genome editing-mediated biofortification. Agronomic biofortification can offer a short-term boost in micronutrients via the application of fertilizers. Biofortification can be accomplished using conventional plant breeding methods, where plants with high levels of vitamins or minerals are crossbred across multiple generations to create new varieties with improved nutritional and agricultural qualities. Transgenic approaches become advantageous when the desired nutrient is not naturally found in the crop or when it is difficult to incorporate adequate bioavailable micronutrients through breeding. Nevertheless, even after developing a transgenic line, it takes several years of conventional breeding to ensure that the transgenes are stably inherited and to incorporate the transgenic traits into varieties favored by farmers. Although transgenic methods may sometimes enhance micronutrient content beyond what is possible with traditional breeding, many countries lack the legal frameworks required for the approval and commercialization of such varieties (Saltzman et al. 2013).

Nevertheless, it is crucial to recognize that biofortification alone cannot fully address or eradicate micronutrient deficiencies across all demographics. While no single strategy can completely resolve the issue of micronutrient malnutrition, biofortification serves as a valuable addition to current measures, offering a sustainable and cost-effective means to supply essential micronutrients by aiming to provide the recommended dietary allowances to the most at-risk populations (Pfeiffer and McClafferty 2007; Qaim et al. 2007). This review explores the biofortification of Solanaceae crops, focusing on potatoes and tomatoes, with essential micronutrients to combat global deficiencies. We discuss various strategies used for enhancing the iron, zinc, iodine, vitamin A, vitamin C, vitamin D, vitamin E, silicon, selenium and flavonoid content in these crops through soil and foliar fertilization, innovative techniques like nanoparticle applications (Table 1), and genetic engineering (Table 2). We also outline strategies and policy recommendations for advancing biofortification to meet global hunger and malnutrition goals.

Biofortification in Solanaceae Crops

Crop biofortification has become a potent method for addressing micronutrient deficiencies. It offers a cost-efficient strategy that ensures a sustainable supply of essential nutrients to impoverished populations. Numerous biofortified crops such as potato, sweet potato, lettuce, tomato, carrots, etc. have been introduced globally (Garg et al.

Table 1 Previous research on agronomic biofortification of solanaceae crops

Biofortification method	Plant	Micronutrient	Reference
Seedling priming alongwith NPK fertilizers	Tomato	Iron	Rabbi et al. (2022)
Seed priming	Tomato	Iron	Ikram et al. (2023)
Foliar fertilization with MNENPK	Eggplant	Iron	Bana et al. (2022)
Root application and foliar spraying	Cherry tomato	Iron	Buturi et al. (2022)
Humic acid coated Fe3O4 nanoparticles	Tomato	Iron	Raiesi-Ardali et al. (2022)
CA/Fe2(SO4)3 and EDTA-Fe·Na treatment	Potato	Iron	Zhang et al. (2022)
Nanocomposite slow-release fertilizer containing chitosan nanographene oxide GO-CS coating and NPKFeZn	Tomato	Iron	Raiesi-Ardali et al. (2024)
Iron-solubilizing rhizobacterial strain inoculum in conjunction with a growth regulator	Potato	Iron	Mushtaq et al. (2024)
Tuber priming	Potato	Zinc	Vergara Carmona et al. 2019)
Foliar and soil application	Potato	Zinc	Kromann et al. (2017)
Foliar fertilization with MNENPK alongwith NPK fertilizers	Eggplant	Zinc	Bana et al. (2022)
Foliar fertilization	Potato	Zinc	White et al. (2017)
Foliar fertilization with Zn, B, Fe, Mn alongwith NPK fertilizers	Potato	Zinc	Moinuddin et al. (2017)
Nano-fertilizer in the form of ZnO-NPs	Cherry tomato	Zinc	Almendros et al. (2022)
Treatment with Zn nanoparticles	Tomato	Zinc	Parra-Torrejón et al. (2023)
Foliar application of zinc oxide NPs	Tomato	Zinc	Ahmed et al. 2023)
Iodine and salicylic acid	Tomato	Iodine	Smoleń et al. 2015)
Potassium iodide application	Pepper	Iodine	Li et al. (2017)
Soil and foliar applications using KIO3	Potato	Iodine	Ledwożyw-Smoleń et al. (2020)
Iodoquinolines and KIO3	Potato	Iodine	Krzemińska et al. (2025)
Foliar application using KIO3	Cherry tomato and mini-plum tomato	Iodine	Somma et al. (2024)
Spraying the whole plant with Se solution	Tomato	Selenium	Shiriaev et al. (2022)
Fertigation and grafting	Cherry tomato	Selenium	Sabatino et al. (2021)
Foliar spraying	Tomato	Silicon	dos Santos et al. (2022)

2018). Since potatoes are a staple in many diets, even small increases in their phytonutrient content can have significant dietary benefits, more so than with foods that are consumed less frequently. In 2022, global potato production reached 375 million tons, out of which 54% is produced by Asian countries (FAOSTAT 2023). Affordability and storability are some key characteristics that increase potato's importance for food security. Tomatoes are also a vital source of nutrition for people across the globe. In 2022, global tomato production reached 186.107 million metric tons, cultivated over 4,917,735 hectares (FAOSTAT 2023). Tomatoes are well-known for their health benefits, which are mainly due to the antioxidant properties of their fruit. This is due to a combination of bioactive compounds, including potassium, lycopene, folate, ascorbic acid, α-tocopherol, and polyphenols (Basu and Imrhan 2007). Hence, potatoes and tomatoes are excellent candidates for biofortification due to their widespread consumption and significant nutritional benefits. Enhancing these crops could substantially improve global nutrient intake and address deficiencies, leveraging their high production volumes and essential health-promoting compounds.

Iron Biofortification

In 2021, global anemia prevalence stood at 24.3%, affecting approximately 1.92 billion people worldwide. Dietary iron deficiency was found to be responsible for 66.2% of all anemia cases (Gardner et al. 2023). Iron deficiency constitutes the leading cause of anemia, with inadequate iron intake responsible for 50% of cases. Extended iron deficiency can result in iron deficiency anemia (IDA), which is estimated

Table 2 Previous research on biofortification of solanaceae crops using transgenic methods

Genes	Plant	Micronutrient	References
GTP cyclohydrolase I	Tomato	Iron	Díaz de la Garza et al. 2004)
aminodeoxychorismate synthase (*AtADCS*)	Tomato	Iron	Díaz de la Garza et al. 2007)
Pteridine, PABA	Potato	Iron	Burgos et al. (2023)
AtNAS1, Arabidopsis nicotianamine synthase 1	Potato	Iron	Zha et al. (2022)
crtB gene	Potato	Vitamin A	Ducreux et al., (2005)
lycopene b-cyclase	Tomato	Vitamin A	Wurbs et al. (2007)
lycopene b-cyclase	Tomato	Vitamin A	Rosati et al. 2000)
bacterial *dxs* gene	Tomato	Vitamin A	Enfissi et al. (2006)
phytoene synthase-1	Tomato	Vitamin A	Fraser et al., (2007)
beta-carotene hydroxylases *CHY1* and *CHY2*	Potato	Vitamin A	Diretto et al. (2007a)
CrtB, *CrtI*, *CrtY*	Potato	Vitamin A	Diretto et al. (2007b)
bacterial carotenoid gene (*crtI*) encoding phytoene desaturase	Tomato	Vitamin A	Römer et al. (2000)
lycopene β-cyclase genes from daffodil	Tomato	Vitamin A	Apel and Bock 2009)
lycopene epsilon cyclase (*LCY-e*)	Potato	Vitamin A	Diretto et al. (2006)
beta-carotene hydroxylase	Potato	Vitamin A	Van Eck et al. (2007)
lycopene b-cyclase (*LCYb*)	Potato	Vitamin A	Song et al. (2016)
Or (*Orange*) gene	Potato	Vitamin A	Lopez et al. (2008)
zeaxanthin epoxidase	Potato	Vitamin A	Römer et al. (2002)
β-carotene ketolase, β-carotene hydroxylase	Tomato	Vitamin A	Huang et al. (2013)
b-Lcy, *b-Chy*	Tomato	Vitamin A	Dharmapuri et al. (2002)
GDP-L-galactose phosphorylase	Tomato, Potato	Vitamin C	Bulley et al. (2012)
GDP-Mannose 3′,5′-epimerase	Tomato	Vitamin C	Zhang et al. (2011)
GDP-mannose pyrophosphorylase	Tomato	Vitamin C	Cronje et al. (2012)
arabinono-1,4-lactone oxidase	Tomato	Vitamin C	Cronje et al. (2012)
GalUR gene	Tomato	Vitamin C	Wevar Oller et al. 2009)
GalUR gene	Potato	Vitamin C	Hemavathi et al. (2010)
GalUR gene	Potato	Vitamin C	Cai et al. (2015)
GLOase gene	Tomato	Vitamin C	Lim et al. (2012)
GMP × GME × GGP × GPP	Tomato	Vitamin C	Li et al. (2019)
SlNL33	Tomato	Vitamin C	Ye et al. (2019)
DHAR1	Potato	Vitamin C	Qin et al. (2011)
DHAR1	Tomato	Vitamin C	Li et al. (2012)
SlZF3, a tomato C2H2 zinc-finger protein coding gene	Tomato	Vitamin C	Li et al. (2017)
7-dehydrocholesterol reductase (*Sl7-DR2*)	Tomato	Vitamin D	Li et al. (2022)
DWARF5A (*SlDWF5A*)	Tomato	Vitamin D	Choi et al. (2023)
synthetic tocopherol operon	Tomato	Vitamin E	Lu et al. (2013)
At-HPPD, *At-HPT*	Potato	Vitamin E	Crowell et al. (2008)
StMGL1 gene	Potato	Methionine	Huang et al. (2014)
threonine synthase	Potato	Methionine	Zeh et al. (2001)
cystathionine γ-synthase (*CGS*)	Potato	Methionine	Di et al. (2003)
Amaranth Albumin 1 (*AmA1*)	Potato	Amino acids and Protein	Chakraborty et al. (2010)
chalcone synthase, chalcone isomerase, and dihydroflavonol reductase	Potato	Flavonoids	Lukaszewicz et al. (2004)
AtMYB12	Tomato, Tobacco	Flavonoids	Luo et al. (2008)
stilbene synthase gene	Tomato	Flavonoids	Giovinazzo et al. (2005)
chalcone isomerase gene	Tomato	Flavonoids	Muir et al. (2001)
HQT	Tomato	Flavonoids	Niggeweg et al. (2004)

to impact about 16.42% of the world's population. (Camaschella 2019; Lopez et al. 2016; Gedfie et al. 2022). IDA is characterized by the World Health Organization (WHO) as having a hemoglobin level below 110 g/L in children aged 6–59 months, below 130 g/L in men, below 120 g/L in women, or below 110 g/L during pregnancy (WHO 2001). One of the nutritional interventions to combat IDA is iron biofortification of staple food crops. The daily iron requirement is 0.71 mg/day for children aged 7–10 years, 1.05 mg/day for adult men, and 1.46 mg/day for adult women (Joint and WHO 2004).

Various strategies, including fertilizer application and genetic engineering, have shown promise in increasing iron content in these crops. Iron biofortification can be accomplished by implementing effective iron fertilizers and optimizing fertilizer management practices on a large scale (Wallace 1988). Zhang et al. (2022) observed that applying citric acid–inorganic iron fertilizer treatment and chelated iron through foliar application had a very pronounced impact on enhancing iron biofortification in potatoes. Rabbi et al. (2022) achieved iron biofortification of 66.28 µg/g in tomato, suggesting that applying iron through seedling priming, followed by a foliar spray during the seedling stage combined with the recommended NPK fertilizers, is an effective approach for iron biofortification in tomatoes. Priming tomato seeds with 10 mg/L iron was shown to improve both the yield and quality of tomato hybrids, particularly increasing fruit Fe content (Ikram et al. 2023). Foliar fertilization with MNENPK during flowering and fruiting stages, in combination with recommended doses of NPK fertilizers led to a significant 26% increase in the iron concentration in eggplant (Bana et al. 2022). Buturi et al. (2022) observed that supplementing cherry tomato plants with 2 mmol/L iron via nutrient solution and 500 umol/L iron through foliar spraying substantially increased iron concentration by up to 190%. This suggests that combining root and foliar applications in soilless systems can facilitate iron biofortification effectively, aiding in combating malnutrition and enhancing tomato quality. Humic acid coated Fe3O4 nanoparticles at a concentration of 50 mg/kg significantly increased plant iron content, providing a more bioavailable iron source for the plants (Raiesi-Ardali et al. 2022). Later, Raiesi-Ardali et al. (2024) synthesized a nanocomposite slow-release fertilizer containing chitosan nanographene oxide GO-CS coating and NPKFeZn, enhancing tomato plant growth by increasing shoot nutrient content. The iron content in the tomato plants was observed to increase by 75% as a result of the slow-release fertilizer treatment. This suggests that humic acid coated Fe_3O_4 nanoparticles can be used as a cost-effective, eco-friendly, and efficient nano-fertilizer to address iron deficiency in plants, highlighting its potential in agricultural applications.

Overexpression of GTP cyclohydrolase I in tomato fruit resulted in a notable increase in pteridine and folate content, with folate levels with folate levels doubling on average compared to the controls. The engineered tomatoes showed enhanced levels of pteridines such as neopterin, monapterin, and hydroxymethylpterin, along with novel pteridine glycosides that had not been previously identified in plants (Díaz de la Garza et al. 2004). In a later study by Díaz de la Garza et al. 2007), they observed that overexpressing aminodeoxychorismate synthase (*AtADCS*) in tomato fruit resulted in an average of 19-fold more PABA content compared to controls, contributing to enhanced folate levels in the fruit. Moreover, the combination of transgenic pteridine and PABA overproduction traits in tomato fruit resulted in a substantial boost in folate accumulation, with vine-ripened fruit exhibiting folate levels up to 25 times higher than those in the controls, providing folate levels that meet the complete adult daily requirement in less than one standard serving. Iron absorption from iron-biofortified yellow-fleshed potatoes was significantly higher compared to non-biofortified potatoes, with total iron absorption being 45.8% higher in the biofortified variety (Burgos et al. 2023). Iron deficiency leads to chlorosis in plants, which can significantly reduce their productivity. Boamponsem et al. (2017) developed Fe-efficient potato plant lines that showed resistance to low iron supply stress compared to iron-inefficient lines and control parental biotype, exhibiting enhanced expression of ferritin in the leaf and iron-regulated transporter in plant roots. Manipulating iron chelation can also prove to be useful in increasing iron content. Transgenic potato plants overexpressing *AtNAS1*, Arabidopsis nicotianamine synthase 1, showed a 2.4-fold higher iron content in their tubers when compared to wild-type tubers (Zha et al. 2022). CRISPR/Cas9-mediated knockdown of the potato transcription factor *StbHLH47* (a homolog of Arabidopsis *POPEYE*) was shown to relieve repression of iron uptake pathways. This resulted in the upregulation of *StNAS4*, *StOPT3*, and *StFRO3*, leading to enhanced iron accumulation in potato tissues (Chauhan et al. 2024). Potato plants, when inoculated with an iron-solubilizing rhizobacterial strain in conjunction with a growth regulator, exhibited enhanced iron absorption along with improved tuber characteristics and agronomic qualities (Mushtaq et al. 2024).

Zinc Biofortification

Crop plants absorb zinc directly from the soil, either as Zn2+ ions or bound to organic acids. This zinc accumulates initially in the roots and is subsequently transported through the xylem to the shoots and leaves. Within plants, zinc serves multiple essential functions, including regulating gene expression, maintaining membrane integrity,

facilitating photosynthesis, protecting against drought and pathogens, and participating in the synthesis of growth-regulating hormones (Noulas et al. 2018). Since zinc cannot be synthesized within the human body, it is necessary to obtain it externally to maintain adequate levels in the body. Zinc is a vital micronutrient crucial for numerous cellular functions such as protein synthesis, cell proliferation, nucleic acid metabolism (including DNA synthesis), and mitosis (Beyersmann and Haase 2001; Maxfield et al. 2022). Ensuring adequate zinc intake is therefore essential for maintaining cellular function and overall health. For an individual to be healthy, the zinc concentration should range from 80 to 120 mcg/dL (Ryu et al. 2020). Zinc levels lower than 70 mcg/dL in women and 74 mcg/dL in men suggest insufficient zinc levels. Maintaining optimal serum zinc levels is crucial for preventing deficiency-related health issues. Research estimates that as much as 17% of the worldwide population is at risk of insufficient zinc intake, with up to 30% of the population in South Asia potentially being deficient. Sub-Saharan Africa and Central America are some other regions that are at risk (Skalny et al. 2021). These statistics underscore the global challenge of zinc deficiency and the need for effective intervention strategies.

Biofortifying potatoes and tomatoes with zinc can help address this issue. Methods such as foliar fertilization and the use of nano-fertilizers have proven effective in enhancing zinc content in these crops. A significant enhancement in zinc bioavailability occurred when potato tubers were immersed for 12 hours in a solution containing 10 mg mL−1 of zinc, making this an effective method for fortification (Vergara Carmona et al. 2019). Such methods offer practical solutions for increasing zinc content in commonly consumed crops. Andean potato cultivars can be agronomically fortified with zinc using foliar and soil-applied zinc fertilizers. Field trials showed that applying high levels of zinc to the leaves increased tuber zinc content by 2.51 times, whereas applying high levels of zinc to the soil resulted in a 1.91-fold increase. Applying zinc fertilizers or zinc-enriched NPK fertilizers provides an expedient solution to increasing zinc concentration in Andean potato tubers (Kromann et al. 2017). Similarly, foliar fertilization with MNENPK during flowering and fruiting stages, in combination with recommended doses of NPK fertilizers resulted in a notable 34% increase in the zinc concentration in eggplant (Bana et al. 2022). Zinc oxide and zinc sulphate proved more effective than zinc nitrate for foliar fertilization in enhancing tuber zinc concentrations while preserving yields. It was observed that even after cooking, the zinc concentration ratio in biofortified tubers compared to non-biofortified ones varied from 2.36 to 3.58 (White et al. 2017).

Moinuddin et al. (2017) observed that applying a micronutrient mixture via foliar spray, which included zinc, boron, iron, and manganese along with NPK, resulted in a 32.01% increase compared to the yield from the control plot. Almendros et al. (2022) discovered that applying zinc as a nano-fertilizer in the form of ZnO nanoparticles can enhance crop yield and achieve sufficient zinc biofortification in cherry tomato crops. Treatment with Zn nanoparticles significantly increased the zinc content in tomatoes by 352% compared to the control and by 65% compared to ZnSO4 treatment. These nanoparticles enabled the tomatoes to supply, per 100 g, up to 10.75% of the daily recommended dietary allowance for zinc for women and 7.82% for men (Parra-Torrejón et al. 2023). Foliar application of zinc oxide nanoparticles increased the uptake efficiency of zinc in tomato plants (Ahmed et al. 2023). Nano-fertilizers represent an innovative approach to improving nutrient delivery and uptake in plants, showcasing the potential of nanotechnology to significantly improve the nutritional quality of crops and address micronutrient deficiencies in human populations.

Iodine Biofortification

Plants have the capability to absorb iodine from the soil, where it exists in both inorganic forms (iodide, I−, and iodate, IO3−) and organic compounds. The speciation and mobility of iodine in soil are influenced by factors such as soil pH, texture, composition, and redox conditions, which in turn affect its availability for root uptake (Medrano-Macías et al. 2016). For children aged 6 and older, sufficient iodine levels are indicated by a median iodine concentration within the range of 100–199 µg/L. For pregnant women, the optimal median iodine concentration ranges from 150 µg/L to 249 µg/L (WHO 2013). Iodine is crucial for the synthesis of thyroid hormones such as triiodothyronine (T3) and tetraiodothyronine (T4). Insufficient iodine intake can lead to a range of health problems known commonly as iodine deficiency disorders (Zimmermann et al. 2008).

Strategies for biofortifying crops with iodine, including soil and foliar applications, can help mitigate this deficiency. Incorporating iodine and SA in the cultivation process positively affected the nutritional value of tomato fruits. The combination of potassium iodide (KI) and salicylic acid (SA) led to a 37% increase in iodine accumulation, while combining KIO3 and SA led to a significant 157% boost in the iodine levels of tomato. These iodine-enriched tomatoes could potentially serve as an alternative dietary source of iodine, contributing to meeting the recommended nutrient intake for adults (Smoleń et al. 2015). Applying iodine in low to moderate doses enhanced the quality of pepper fruits by increasing their ascorbic acid and soluble sugar levels. Pepper fruits cultivated in 0.25 to 5.0 mg/L KI contained iodine levels that met the recommended dietary intake of

150 g/day for adults. Incorporating iodine led to increased chlorophyll-a concentration, catalase, peroxidase, superoxide dismutase activities, and decreased malondialdehyde concentration in pepper plants, enhancing photosynthetic and antioxidant capacities for improved fruit quality (Li et al. 2017).

Dobosy et al. (2020) deemed the potato plant as unsuitable for iodine biofortification as the iodine levels did not increase significantly when irrigated with KI doped water. However, Ledwożyw-Smoleń et al. (2020) suggested that biofortified potatoes could serve as an extra source of iodine in everyday diets, addressing iodine deficiency concerns in various populations. Applying iodine to potato tubers through soil and foliar applications was found effective in increasing the iodine content without negatively impacting yield, dry matter, starch, or sugar content. Foliar spraying with 2.0 kg/ha KIO3 was observed to be the most effective method for iodine biofortification of potatoes. The iodine levels achieved in the biofortified potatoes could potentially contribute up to 25% of the RDA for iodine, highlighting their potential as a major source of this crucial nutrient. Krzemińska et al. (2025) studied that iodoquinolines were successfully absorbed, transported, and accumulated in potato plants, impacting the iodine enrichment levels in the tubers. KIO3 demonstrated the highest iodine biofortification levels in potato tubers, surpassing the iodoquinoline compounds tested. Different iodoquinolines showed varying efficacy in enriching tubers with iodine, with 5,7-diiodo-8-quinolinol being one of the most effective after KIO3. Higher doses (50 µM) of iodoquinolines resulted in greater iodine enrichment of the tubers compared to lower doses (10 µM). Applying a 1 mM KIO3 solution to mini-plum and cherry tomatoes grown in a soilless system proved to be a successful method for iodine biofortification without adversely affecting plant health or yield. Consuming 100 g of raw tomatoes from both varieties could provide a significant portion of the recommended dietary allowance (Somma et al. 2024).

Vitamin A/Carotenoids Biofortification

Vitamin A is essential for cellular communication, immune system performance, growth, development, and reproductive health in both men and women. It aids in cell growth and differentiation, which is vital for the correct development of the heart, lungs and eyes (BPMarriott 2020; Ross and Vitamin 2014; IMFNB 2000). The human diet provides vitamin A from two sources: preformed vitamin A and provitamin A carotenoids. Provitamin A carotenoids, which include beta-carotene, alpha-carotene, and beta-cryptoxanthin, are plant pigments. In the intestine, the enzyme beta-carotene monooxygenase type 1 transforms these carotenoids into vitamin A (IMFNB 2001; Kedishvili 2013). Globally, the majority of vitamin A deficiency cases are found in children under the age of 5 in developing countries. While the global estimates of vitamin A deficiency in young children are declining, approximately 30% of children under 5 are still affected, and this deficiency is responsible for about 2% of all deaths in this age group (Wirth et al. 2017). A retinol level under 0.70 µmol/L indicates vitamin A deficiency in both children and adults, whereas a level below 0.35 µmol/L suggests a severe deficiency. (WHO 2009).

To address vitamin A deficiency, biofortification of potatoes has been a focus of several studies. Genetic modifications have successfully increased vitamin A content in both potatoes and tomatoes. Transgenic potato plants expressing the *Erwinia uredovora crtB* gene showed a significant increase in carotenoid content in tubers, with levels reaching up to 78 µg carotenoid/g dry weight (Ducreux et al. 2005). Plastid expression of the *Erwinia herbicola* lycopene b-cyclase (*crtY*) gene enhanced carotenoid synthesis in tomato by four-fold by promoting the conversion of lycopene to beta-carotene. The transplastomic plants showed increased pro-vitamin A content and herbicide resistance (Wurbs et al. 2007). Additionally, Arabidopsis lycopene b-cyclase over-expressing transformants showed increased beta-carotene content and altered fruit color in tomato plants. Lycopene levels were also enhanced with pleiotropic phenotypes observed in some transformants. High lycopene and beta-carotene content in transgenic tomato might prove beneficial for dietary antioxidant intake (Rosati et al. 2000). Genetically manipulating the methylerythritol-4-phosphate (MEP) pathway by overexpressing the bacterial 1-deoxy-D-xylulose-5-phosphate synthase (dxs) gene resulted in a 1.6-fold enhancement in carotenoid levels in transgenic tomatoes. The most significant increases were observed in phytoene by 2.4-fold and β-carotene by 2.2-fold (Enfissi et al. 2006). Overexpression of phytoene synthase-1 (PSY-1) in tomato resulted in a 1.5- to 3.0-fold elevation in phytoene, lycopene, and β-carotene in some transgenic lines (Fraser et al. 2007).

Diretto et al. (2007a) found that inhibiting the beta-carotene hydroxylases CHY1 and CHY2 in potato tubers led to notable increases in both total carotenoids and beta-carotene levels, with beta-carotene rising by as much as 38 times and total carotenoids increasing up to 4.5 times. In another study, Diretto et al. 2007b) introduced a bacterial mini-pathway into potatoes to produce beta-carotene (Provitamin A) by incorporating three Erwinia genes (CrtB, CrtI, and CrtY) regulated by tuber-specific promoters. Expressing all three genes specifically in tubers produced deep yellow ("golden") tubers with a significant increase in carotenoids, 20 times higher, reaching 114 µg/g dry weight and a 3600-fold increase in beta-carotene, totaling 47 µg/g dry weight.

A 250 g serving of these golden potatoes can provide 50% of the RDA for Vitamin A. To enhance carotenoid levels and composition in tomato fruit, Römer et al. (2000) created transgenic lines incorporating a bacterial carotenoid gene (crtI) that encodes phytoene desaturase, which converts phytoene into lycopene. This alteration led to a threefold rise in β-carotene, which represented about 45% of the total carotenoid content. Apel and Bock (2009) studied the incorporation of lycopene β-cyclase genes from daffodil into the tomato plastid genome, which enabled the effective transformation of lycopene into provitamin A (β-carotene) in tomato fruits. The developed transplastomic tomatoes exhibited over a 50% increase in overall carotenoid content, suggesting an improved flux through the pathway in chromoplasts.

Diretto et al. (2006) silenced the lycopene epsilon cyclase (LCY-e) gene, which is crucial for carotenoid biosynthesis, in potato tubers. This intervention led to up to a 14-fold increase in beta-carotene levels and up to a 2.5-fold increase in total carotenoid levels. Van Eck et al. (2007) utilized RNA interference to silence the beta-carotene hydroxylase gene, responsible for converting beta-carotene to zeaxanthin. The resulting transgenic lines exhibited increased beta-carotene levels in tubers, reaching up to 331 µg per 100 g of fresh weight. This amount could provide approximately 18% of the RDA for vitamin A for children aged one to three, who are especially susceptible to vitamin A deficiency-related health issues. Song et al. (2016) overexpressed lycopene β-cyclase (LCYb) in potato plants, which is involved in the synthetic pathway of carotenoids from lycopene, resulting in a 1.5 to 1.9-fold increase in β-carotene content in tubers. The Or (Orange) gene was discovered as a new mutation that controls the accumulation of carotenoids in plants. Overexpression of the Or gene in potatoes led to a tenfold enhancement in carotenoid content in the transgenic tubers (Lopez et al. 2008). Römer et al. (2002) inhibited the conversion of zeaxanthin to violaxanthin using zeaxanthin epoxidase in potato plants, resulting in up to a 130-fold increase in zeaxanthin levels (40 µg/g dry weight) in transgenic tubers. These tubers also exhibited up to a 5.7-fold increase in total carotenoid content and a 2 to 3-fold increase in α-tocopherol levels. Huang et al. (2013) employed co-expression of *Chlamydomonas reinhardtii* β-carotene ketolase and *Haematococcus pluvialis* β-carotene hydroxylase to convert carotenes into astaxanthin in tomatoes. This resulted in a substantial increase in the concentration of free astaxanthin in the leaves, increasing the total carotenoids in transgenic tomato plants by up to 16-fold. Additionally, overexpression of lycopene β-cyclase (b-Lcy) and β-carotene hydroxylase (b-Chy) resulted in significant accumulation of xanthophylls such as L-cryptoxanthin and zeaxanthin in transgenic tomatoes (Dharmapuri et al. 2002).

Vitamin C Biofortification

Vitamin C is a water-soluble vitamin and antioxidant that plays a crucial role as a co-factor in the synthesis of collagen, the metabolism of carnitine and catecholamines, and the absorption of dietary iron. As humans are unable to synthesize vitamin C, it must be obtained from dietary sources such as fruits and vegetables. Rich sources of vitamin C include citrus fruits, berries, tomatoes, potatoes, and green leafy vegetables (Fenech et al. 2018). Recognizing the importance of vitamin C for health, researchers have focused on increasing its levels in staple crops through biofortification.

Vitamin C serves as a vital antioxidant and performs numerous roles in maintaining optimal health and preventing diseases (Carr and Frei 1999a; Carr and Frei 1999b). Severe vitamin C deficiency leads to scurvy, which appears 2–3 months after a diet devoid of vitamin C. It is marked by numerous hemorrhages and, if not treated, can be fatal (Ceglie et al. 2019). In adult males, the body content of ascorbate reaches around 20 mg/kg, or 1500 mg, at saturation. Clinical symptoms of scurvy appear when the body content drops below 300–400 mg, and these symptoms completely vanish when the body content reaches approximately 1000 mg. Enhancing the vitamin C content in potatoes and tomatoes through biofortification can improve public health outcomes, particularly in regions where fresh fruit and vegetable consumption is low.

One effective strategy has been the over-expression of the L-galactose pathway gene GDP-L-galactose phosphorylase (GGP), which significantly increased ascorbate concentration in commercially important edible crops like tomato and potato (Bulley et al. 2012). Additionally, regulating endogenous GDP-Mannose 3′,5′-epimerase (GME) in tomato has shown potential to boost AsA biosynthesis. Zhang et al. (2011) observed that ascorbic acid levels can be enhanced by overexpressing the two GME (GDP-Mannose 3′,5′-epimerase) gene family members (*SlGME1* and *SlGME2*) in tomato. They found that in ascorbic acid content increased by 60% and 24% in GME1- and GME2-overexpressing tomato plants, respectively. Manipulating L-ascorbic acid biosynthesis pathways, particularly the Smirnoff–Wheeler pathway, has shown promising results in increasing ascorbate (AsA) levels in tomato plants. Cronje et al. (2012) found that overexpression of GDP-mannose pyrophosphorylase (GMP) in S. lycopersicum led to up to a 70% increase in ascorbic acid levels in leaves, 50% in green fruit, and 35% in red fruit. Additionally, expression of arabinono-1,4-lactone oxidase (ALO) significantly redirected biosynthetic flux towards ascorbic acid by up to 54%, further supporting the feasibility of manipulating AsA levels in plants.

Wevar Oller et al. (2009) observed that tomato hairy roots could biosynthesize L-ascorbic acid in situ and de novo, accumulating the vitamin independently of growth. Both wild-type and transgenic tomato hairy roots showed an increase in total ascorbic acid content when supplied with D-galactose or D-galacturonic acid, indicating the operation of the galacturonate pathway for vitamin C biosynthesis. The transgenic hairy root clone overexpressing the *GalUR* gene exhibited a remarkable 2.5-fold elevation in ascorbic acid content when treated with D-galacturonic acid, suggesting the effectiveness of the D-galacturonic acid precursor in enhancing vitamin C production. In addition, potato plants overexpressing the strawberry GalUR gene (D-galacturonic acid reductase) led to a 1.6 to 2-fold increase in the ascorbic acid contents of transgenic tubers. This increase can be attributed to enhanced mRNA expression and activity of the GalUR enzyme (Hemavathi et al. 2010). Similarly, overexpression of the rat L-gulono-γ-lactone oxidase (GLOase) gene in tomato plants led to a 1.5-fold increase in ascorbic acid (AsA) content in fruits, correlating with enhanced GLOase activity and mRNA expression (Lim et al. 2012). Effective biofortification to increase ascorbic acid content in tomatoes has been achieved by pyramiding GMP × GME, GGP × GPP (L-Gal-1-P phosphatase), and the tetravalent combination of GMP × GME × GGP × GPP (Li et al. 2019). The reduced expression of *SlNL33*, the gene encoding an NL type NBS-LRR resistance protein, increased AsA accumulation in tomato plants by upregulating key genes involved in the redox pathway (Ye et al. 2019). Overexpression of *DHAR1* led to increased dehydro-ascorbate reductase (DHAR) activity and subsequently increased AsA content in leaves and tuber of transgenic potato plants (Qin et al. 2011). Similarly, overexpression of DHAR1 led to increased ascorbic acid content in both leaves and fruits in tomato when compared to wild type plants (Li et al. 2012). Li et al. (2018) observed that the overexpression of SlZF3, a tomato C2H2 zinc-finger protein coding gene, led to increased accumulation of AsA in tomato plants by interacting with CSN5B, a COP9 signalosome component. These studies collectively demonstrate the potential of genetic engineering to biofortify vitamin C content in important crops, thereby enhancing their nutritional value and contributing to better health outcomes.

Other Micronutrients

Vitamin D: Vitamin D is a fat-soluble vitamin that occurs naturally in certain foods and is also available as a dietary supplement. Vitamin D, derived from foods, sun exposure and supplements, is biologically inactive and requires two hydroxylation processes in the body to become active. The average RDA for vitamin D is 15 µg/day (IMFNB 2010). In children, a lack of vitamin D leads to rickets, a condition where bone tissue fails to mineralize correctly, causing soft bones and skeletal deformities (Elder and Bishop 2014). Li et al. (2022) utilized CRISPR/Cas9 genome editing to silence 7-dehydrocholesterol reductase (Sl7-DR2) in tomato plants, resulting in an elevated accumulation of 7-dehydrocholesterol (7-DHC), also referred to as provitamin D3. After exposure to UVB light, the mutant tomato plants exhibited vitamin D3 levels reaching up to 30% of the RDA. As the leaves of the mutant plant were also a rich source of vitamin D3, the waste material from these biofortified tomatoes could be utilized for vitamin D supplement production, adding an economic and health benefit to the biofortification strategy. Choi et al. (2023) observed that CRISPR/Cas9 genome-editing of tomato DWARF5A gene (*SlDWF5A*), a 7-DHC reductase homolog, led to increased accumulation of provitaminD3 in tomato fruits of up to 6 µg/g dry weight. These studies point to the possibility of increasing the accumulation of provitamin D3 in Solanaceae crops such as potato, eggplant, pepper, etc. through genetic modifications in phytosterol biosynthesis pathway.

Vitamin E: Vitamin E, which includes eight biologically active tocopherols, plays a crucial role as an antioxidant in the health and well-being of premature newborns (Thibeault 2000). By scavenging free radicals, it helps reduce the risk of lipid peroxidation and oxidant-related harm in preterm infants (Roberts and Knight 1987). However, achieving these protective effects requires higher serum levels of tocopherols than what can be attained through standard oral doses of vitamin E. Notably, recent research sheds light on enhancing vitamin E levels in plants. Transplastomic plants expressing the synthetic tocopherol operon exhibited up to a 10-fold increase in vitamin E (tocochromanol) levels in tobacco and tomato. In tobacco, leaves showed superior enhancement compared to fruits, with the synthetic operon expression leading to more efficient conversion of intermediates and a notable increase in tocopherol content (Lu et al. 2013). Similarly, constitutive overexpression of two Arabidopsis transgenes, At-HPPD (p-hydroxyphenylpyruvate dioxygenase) and At-HPT (homogentisate phytyltransferase) in potato led to an increase of up to 266% and 106% in tuber α-tocopherol levels. However, the accumulation of α-tocopherol was less in tubers compared to leaves and seeds (Crowell et al. 2008).

Selenium and Silicon: Selenium is important for antioxidant defense and thyroid function. Biofortifying crops with selenium can help address deficiencies, particularly in regions with selenium-poor soils. Adding sodium selenite to the nutrient solution significantly biofortified tomatoes by increasing selenium content across various components (fruits, leaves, and stems). The 5 mg L−1 treatment produced up to twice the selenium concentration in fruits compared to

the control (Rahim et al. 2020). Selenium enriched tomatoes can also be produced effectively by spraying the entire plant with a selenium solution, leading to selenium accumulation in the fruit mainly through direct penetration (Shiriaev et al. 2022). Enriching cherry tomatoes with selenium and using grafting techniques significantly improved selenium accumulation (Sabatino et al. 2021). Applying silicon via foliar spraying during the reproductive stage of tomato plants successfully enhances the silicon content in the fruit. Silicic acid, alkaline silicate, and nanosilica have been observed to enhance the silicon and ascorbic acid levels (dos Santos et al. 2022).

Amino Acids and Protein: Amino acids, as organic compounds, primarily consist of nitrogen, carbon, hydrogen, and oxygen. They are essential for the growth and proper function of living organisms. In total, there are 20 different amino acids, each with distinct properties. While all 20 contribute to health, only 9 are considered as essential as our bodies cannot synthesize them internally and hence, they must be acquired through our diet. Research into plant genetics has explored ways to enhance amino acid and protein content to improve nutritional value. For instance, in potato tubers, silencing the *StMGL1* gene led to an increased methionine-to-isoleucine ratio in the free amino acid profile, indicating the role of methionine gamma-lyase (MGL) in limiting methionine accumulation (Huang et al. 2014). Antisense inhibition of threonine synthase in potato plants led to a significant increase of up to 239-fold in methionine levels without affecting cystathionine gamma-synthase activity (Zeh et al. 2001). Di et al. (2003) observed that increasing soluble methionine levels in potato via Arabidopsis cystathionine γ-synthase (CGS) overexpression led to up to 6-fold higher methionine levels, resulting in 2.4–4.4.4.4-fold increased methional levels in processed potatoes. The processed potatoes containing increased methionine exhibited enhanced flavor stability. Protein-rich potato expressing a seed protein, Amaranth Albumin 1 (AmA1), was developed from proteome rebalancing in transgenic tubers, leading to a 60% increase in protein content. These transgenic potatoes exhibited higher levels of essential amino acids, which improved photosynthesis and biomass production. AmA1 potato tubers were also nontoxic and nonallergenic (Chakraborty et al. 2010).

Flavonoids: Flavonoids are a broad category of bioactive compounds, characterized as polyphenolic, and widely found as plant secondary metabolites, with fruits and vegetables being their most abundant sources (Corradini et al. 2011). Owing to their strong antioxidant and antiradical properties, flavonoids help the human body significantly in fighting various diseases (Grassi et al. 2010; Ren et al. 2003; Gutierrez-Merino et al. 2011; Liu et al. 2021). To enhance the nutritional value of crops, various genetic engineering methods have been developed to biofortify flavonoids. Overexpressing chalcone synthase (CHS), chalcone isomerase (CHI), and dihydroflavonol reductase (DFR) genes in potato tubers increased phenolic compounds, specifically flavonoids, enhancing antioxidant capacity (Lukaszewicz et al. 2004). AtMYB12, a flavonol-specific transcriptional activator, was found to induce high flavonol and caffeoyl quinic acid (CQA) levels in tomato and tobacco. AtMYB12 enhanced total flavonols and CQAs in tomato fruit by 70-fold and 20-fold, respectively. The antioxidant capacity was also increased by up to five-fold in tomato fruit (Luo et al. 2008). Tomato plants that expressed the grapevine stilbene synthase gene showed increased levels of resveratrol. This enhancement not only improved the plants' overall antioxidant capacity but also boosted the content of ascorbate and glutathione in the edible tissues, thereby improving the nutritional contents of the tomato fruit (Giovinazzo et al. 2005). Transgenic tomato lines overexpressing the Petunia chalcone isomerase gene demonstrated up to a 78-fold increase in flavonols, particularly rutin, compared to control lines. Processed products from these high-flavonol tomatoes retained 65% of the flavonols found in the fresh fruit, indicating their potential as ingredients for functional foods (Muir et al. 2001). Additionally, overexpression of the hydroxycinnamoyl-CoA quinate: hydroxycinnamoyl transferase (HQT) enzyme in tomatoes resulted in higher levels of chlorogenic acid (CGA), thereby enhancing their antioxidant capacity (Niggeweg et al. 2004).

Addressing Future Challenges in Biofortification: Strategies and Policy Recommendations

The United Nations' Sustainable Development Goal 2 (UN-SDG2) seeks to eliminate all types of hunger and malnutrition by 2030. Achieving this ambitious target necessitates the execution of biofortified crops as varieties developed through traditional breeding methods are usually more accepted by consumers compared to those created through genetic engineering, which have encountered more resistance. A key requirement for an effective biofortification strategy is the capability to identify and cultivate local staple crop varieties that are rich in essential micronutrients that are scarce in the local diet. Advances in gene cloning and genome editing technologies have accelerated the development of novel crops, providing the means to minimize trait loss due to unstable genomic contexts and optimize gene expression levels. The future of biofortification in Solanaceae crops holds significant promise, contingent on overcoming several key challenges and embracing emerging opportunities. However, several challenges such as ethical

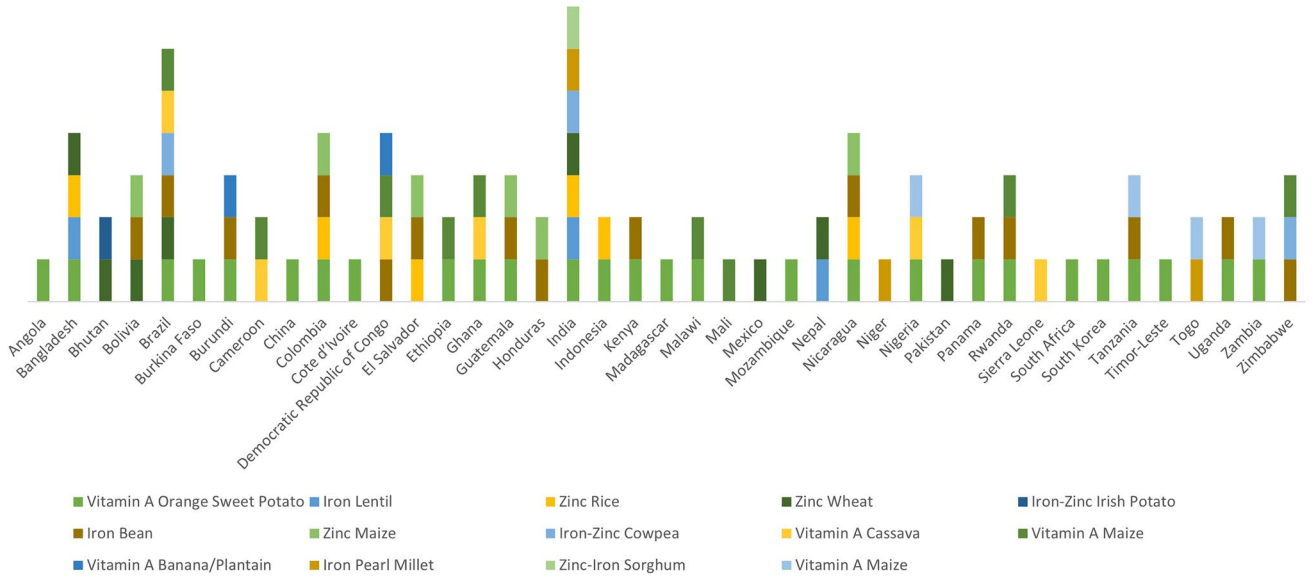


Fig 2 HarvestPlus biofortified crops available in countries around the world (Data from HarvestPlus and International Potato Center)

concerns regarding biofortified transgenic potatoes must be addressed to fully realize the potential of biofortification. One primary challenge is addressing public skepticism regarding genetically modified crops. Despite endorsements from scientific communities, public acceptance remains low. Transparent communication and independent, science-based information dissemination are crucial to gain public trust and acceptance of biofortified crops. It's crucial to develop and update regulatory frameworks for genetically modified foods and crops. The shift from process-based regulation to trait-focused approaches, should be adopted globally. Over the past decade, substantial progress has been achieved in breeding nutrient-rich varieties of many staple food crops, aided by institutions such as HarvestPlus and the International Potato Center as seen in Fig. 2. Ensuring the accessibility of biofortified crops to vulnerable populations and smallholder farmers is crucial. Demonstrating clear health and economic benefits will be essential to gain broader acceptance and adoption. Involving local agricultural institutions in field trials and development phases, particularly in regulatory-benign countries, is crucial for crops requiring extensive breeding programs.

Integrating genetic and agronomic biofortification methods will likely yield the most effective results. While genetic biofortification enhances the plant's natural ability to assimilate micronutrients, agronomic biofortification, through advanced fertilization techniques, ensures sustained nutrient availability in the soil. Innovations in agronomic biofortification, such as foliar application and nutri-priming, have shown increased nutrient use efficiency. Future efforts should focus on an integrated approach that combines varietal improvement, soil conditioning, and advanced agronomic techniques. This holistic strategy can make biofortification a pivotal tool in combating malnutrition and global food insecurity. Continued research and development, along with coordinated efforts to overcome existing challenges, are essential to fully realize the potential of biofortified Solanaceae crops in improving global nutrition and health outcomes.

Conclusions

Biofortification of Solanaceae crops, particularly potatoes and tomatoes, is a promising strategy to combat global micronutrient deficiencies. Advances in breeding, genetic engineering, and agronomic practices have significantly increased the nutritional profiles of these staples, enhancing their vitamin A, vitamin C, iron, and zinc content. These improvements can notably improve the nutritional status of vulnerable populations. Challenges remain, such as public skepticism towards genetically modified crops and the need for updated regulatory frameworks. Transparent communication and science-based information are essential to gain public trust. Ensuring that biofortified crops reach smallholder farmers and vulnerable populations is crucial for widespread adoption. Future efforts should integrate genetic and agronomic biofortification methods to maximize nutrient use efficiency and impact. Combining varietal improvement with advanced agronomic techniques can make biofortification a key strategy in addressing malnutrition and global food insecurity. Coordinated efforts to overcome these challenges are essential to fully realize the potential of

biofortified Solanaceae crops in improving global nutrition and health outcomes.

Acknowledgments The authors are thankful to the Department of Biotechnology, Panjab University, Chandigarh for providing research facilities. Aiana is grateful to the University grants commission (U.G.C.), New Delhi for providing the junior and senior research fellowship.

Author Contributions KS - Conceptualization; Funding acquisition; Investigation; Methodology; Project administration; Supervision; Resources. Aiana- Data curation; Formal analysis; Validation; Visualization; Writing - original draft; and Writing - review & editing. SKU - Writing - review & editing; Formal analysis; Validation.

Funding Authors are thankful to Department of Biotechnology, Government of India and SPARC, MoE, GoI, for the research grant to Prof Kashmir Singh.

Data Availability The authors confirm that the data supporting the findings of this study are available within the article.

Declarations

Conflict of Interest No potential conflict of interest was reported by the authors.

References

Ahmed R, Uddin MK, Quddus MA, Samad MY, Hossain MM, Haque AN (2023) Impact of foliar application of zinc and zinc oxide nanoparticles on growth, yield, nutrient uptake and quality of tomato. Horticulturae 9(2):162

Almendros P, González D, Fernández MD, García-Gomez C, Obrador A (2022) Both Zn biofortification and nutrient distribution pattern in cherry tomato plants are influenced by the application of ZnO nanofertilizer. Heliyon. https://doi.org/10.1016/j.heliyon.2022.e09130

Apel W, Bock R (2009) Enhancement of carotenoid biosynthesis in transplastomic tomatoes by induced lycopene-to-provitamin A conversion. Plant Physiol 151(1):59–66

Bana RS, Jat GS, Grover M, Bamboriya SD, Singh D, Bansal R et al (2022) Foliar nutrient supplementation with micronutrient-embedded fertilizer increases biofortification, soil biological activity and productivity of eggplant. Sci Rep 12(1):5146. https://doi.org/10.1038/s41598-022-09247-0. (**PMID: 35338233**)

Basu A, Imrhan V (2007) Tomatoes versus lycopene in oxidative stress and carcinogenesis: conclusions from clinical trials. Eur J Clin Nutr 61(3):295–303. https://doi.org/10.1038/sj.ejcn.1602510. (**PMID: 16929242**)

Beyersmann D, Haase H (2001) Functions of zinc in signaling, proliferation and differentiation of mammalian cells. BioMetals 14(3–4):331–341. https://doi.org/10.1023/a:1012905406548. (**PMID: 11831463**)

Boamponsem GA, Leung DWM, Lister C (2017) Insights into resistance to Fe deficiency stress from a comparative study of in vitro-selected novel Fe-efficient and Fe-inefficient potato plants. Front Plant Sci 8:1581. https://doi.org/10.3389/fpls.2017.01581. (**PMID: 28955367**)

Bulley S, Wright M, Rommens C, Yan H, Rassam M, Lin-Wang K et al (2012) Enhancing ascorbate in fruits and tubers through over-expression of the L-galactose pathway gene GDP-L-galactose phosphorylase. Plant Biotechnol J 10(4):390–397. https://doi.org/10.1111/j.1467-7652.2011.00668.x

Burgos G, Liria R, Zeder C, Kroon PA, Hareau G, Penny M et al (2023) Total iron absorbed from iron-biofortified potatoes is higher than that from nonbiofortified potatoes: a randomized trial using stable iron isotopes in women from the Peruvian Highlands. J Nutr 153(6):1710–1717. https://doi.org/10.1016/j.tjnut.2023.04.010

Buturi CV, Coelho SRM, Cannata C, Basile F, Giuffrida F, Leonardi C et al (2022) Iron biofortification of greenhouse cherry tomatoes grown in a soilless system. Horticulturae 8(10):858. https://doi.org/10.3390/horticulturae8100858

Cai X, Zhang C, Ye J, Hu T, Ye Z, Li H et al (2015) Ectopic expression of FaGalUR leads to ascorbate accumulation with enhanced oxidative stress, cold, and salt tolerance in tomato. Plant Growth Regul 76(2):187–197. https://doi.org/10.1007/s10725-014-9988-7

Camaschella C (2019) Iron deficiency. Blood 133(1):30–39. https://doi.org/10.1182/blood-2018-05-815944. (**PMID: 30401704**)

Carr A, Frei B (1999) Does vitamin C act as a pro-oxidant under physiological conditions? FASEB J 13(9):1007–1024. https://doi.org/10.1096/fasebj.13.9.1007

Carr AC, Frei B (1999) Toward a new recommended dietary allowance for vitamin C based on antioxidant and health effects in humans. Am J Clin Nutr 69(6):1086–1107. https://doi.org/10.1093/ajcn/69.6.1086

Ceglie G, Macchiarulo G, Marchili MR, Marchesi A, Rotondi Aufiero LR, Di Camillo C et al (2019) Scurvy: still a threat in the well-fed first world? Arch Dis Child 104(4):381–383. https://doi.org/10.1136/archdischild-2018-315496

Chakraborty S, Chakraborty N, Agrawal L, Ghosh S, Narula K, Shekhar S, Naik PS, Pande PC, Chakrborti SK, Datta A (2010) Next-generation protein-rich potato expressing the seed protein gene AmA1 is a result of proteome rebalancing in transgenic tuber. Proc Natl Acad Sci 107(41):17533–17538. https://doi.org/10.1073/pnas.1006265107

Chauhan H, Alok A, Upadhyay SK, Pandey A, Singh K (2024) CRISPR/Cas9 edited StbHLH47 lines exhibit altered expression profiling of iron regulating genes and increased iron content in *Solanum tuberosum*. Curr Plant Biol 38(1):100354

Choi S, You MK, Jeon YA, Lee J, Kim J, Park YJ et al (2023) Metabolic engineering to enhance provitamin D3 accumulation in edible tomatoes. GEN Biotechnology 2(3):219–227. https://doi.org/10.1089/genbio.2023.0015

Corradini E, Foglia P, Giansanti P, Gubbiotti R, Samperi R, Laganà A (2011) Flavonoids: chemical properties and analytical methodologies of identification and quantitation in foods and plants. Nat Prod Res 25(5):469–495. https://doi.org/10.1080/14786419.2010.482054

Cronje C, George GM, Fernie AR, Bekker J, Kossmann J, Bauer R (2012) Manipulation of l-ascorbic acid biosynthesis pathways in Solanum lycopersicum: elevated GDP-mannose pyrophosphorylase activity enhances l-ascorbate levels in red fruit. Planta 235(3):553–564. https://doi.org/10.1007/s00425-011-1525-6

Crowell EF, McGrath JM, Douches DS (2008) Accumulation of vitamin E in potato (Solanum tuberosum) tubers. Transgenic Res 17(2):205–217. https://doi.org/10.1007/s11248-007-9091-1

Desa UN (2023) The sustainable development goals report 2023. special ed. UN DESA. © UN DESA. Available from: https://unstats.un.org/sdgs/report/2023/

de la Díaz Garza RD, Quinlivan EP, Klaus SMJ, Basset GJC, Gregory JF, Hanson AD (2004) Folate biofortification in tomatoes by engineering the pteridine branch of folate synthesis. Proc Natl Acad Sci USA 101(38):13720–13725. https://doi.org/10.1073/pnas.0404208101. (**PMID: 15365185**)

de la Díaz Garza RID, Gregory JF, Hanson AD (2007) Folate biofortification of tomato fruit. Proc Natl Acad Sci U S A 104(10):4218–4222. https://doi.org/10.1073/pnas.0700409104. (**PMID: 17360503**)

Dharmapuri S, Rosati C, Pallara P, Aquilani R, Bouvier F, Camara B, Giuliano G (2002) Metabolic engineering of xanthophyll content in tomato fruits. FEBS Lett 519(1-3):30–34. https://doi.org/10.1016/S0014-5793(02)02699-6

Di R, Kim J, Martin MN, Leustek T, Jhoo J, Ho CT, Tumer NE (2003) Enhancement of the Primary Flavor Compound Methional in Potato by Increasing the Level of Soluble Methionine. J Agric Food Chem 51(19):5695–5702. https://doi.org/10.1021/jf030148c

Diretto G, Tavazza R, Welsch R, Pizzichini D, Mourgues F, Papacchioli V, Beyer P, Giuliano G (2006) Metabolic engineering of potato tuber carotenoids through tuber-specific silencing of lycopene epsilon cyclase. BMC Plant Biol 6(1):13

Diretto G, Welsch R, Tavazza R, Mourgues F, Pizzichini D, Beyer P, Giuliano G (2007a) Silencing of beta-carotene hydroxylase increases total carotenoid and beta-carotene levels in potato tubers. BMC Plant Biol 7(1):11

Diretto G, Al-Babili S, Tavazza R, Papacchioli V, Beyer P, Giuliano G (2007b) Metabolic engineering of potato carotenoid content through tuber-specific overexpression of a bacterial mini-pathway. PLoS One 2(4):e350

Dobosy P, Endrédi A, Sandil S, Vetési V, Rékási M, Takács T, Záray G (2020) Biofortification of potato and carrot with iodine by applying different soils and irrigation with iodine-containing water. Front Plant Sci 9(11):593047

dos Santos MM, da Silva GP, Prado RD, Pinsetta Junior JS, Mattiuz BH, Braun H (2022) Biofortification of tomato with stabilized alkaline silicate and silicic acid, nanosilica, and potassium silicate via leaf increased ascorbic acid content and fruit firmness. J Plant Nutr 45(6):896–903. https://doi.org/10.1080/01904167.2021.1994599

Ducreux LJ, Morris WL, Hedley PE, Shepherd T, Davies HV, Millam S, Taylor MA (2005) Metabolic engineering of high carotenoid potato tubers containing enhanced levels of β-carotene and lutein. J Exp Bot 56(409):81–9

Elder CJ, Bishop NJ (2014) Rickets. Lancet 383(9929):1665–1676. https://doi.org/10.1016/S0140-6736(13)61650-5

Enfissi EM, Fraser PD, Bramley PM (2006) Genetic engineering of carotenoid formation in tomato. Phytochem Rev 5(1):59–65

FAO (2023a) Progress report on follow-up to the second international conference on nutrition (icn2), including implementation of the United Nations decade of action on nutrition. Food and Agriculture Organization.

Fao IFAD (2023b) The state of food security and nutrition in the world 2023. In: UNICEF; WFP; WHO; eBooks. https://doi.org/10.4060/cc3017en

FAOSTAT (2023) World food and agriculture – statistical yearbook 2023; 2023. Available from: https://openknowledge.fao.org/handle/20.500.14283/cc8166en

Fenech M, Amaya I, Valpuesta V, Botella MA (2018) Vitamin C content in fruits: biosynthesis and regulation. Front Plant Sci 9:2006. https://doi.org/10.3389/fpls.2018.02006

Fraser PD, Enfissi EM, Halket JM, Truesdale MR, Yu D, Gerrish C, Bramley PM (2007) Manipulation of phytoene levels in tomato fruit: effects on isoprenoids, plastids, and intermediary metabolism. Plant Cell 19(10):3194–211

Gardner WM, Razo C, McHugh TA, Hagins H, Vilchis-Tella VM, Hennessy C, Taylor HJ, Perumal N, Fuller K, Cercy KM, Zoeckler LZ (2023) Prevalence, years lived with disability, and trends in anaemia burden by severity and cause, 1990–2021: findings from the Global Burden of Disease Study 2021. Lancet Haematol 10(9):e713–e734. https://doi.org/10.1016/S2352-3026(23)00160-6

Garg M, Sharma N, Sharma S, Kapoor P, Kumar A, Chunduri V et al (2018) Biofortified crops generated by breeding, agronomy, and transgenic approaches are improving lives of millions of people around the world. Front Nutr 5:12. https://doi.org/10.3389/fnut.2018.00012

Gedfie S, Getawa S, Melku M (2022) Prevalence and associated factors of iron deficiency and iron deficiency anemia among under-5 children: a systematic review and meta-analysis. Glob Pediatr Health 9:2333794X221110860. https://doi.org/10.1177/2333794X221110860

Giovinazzo G, D'Amico L, Paradiso A, Bollini R, Sparvoli F, DeGara L (2005) Antioxidant metabolite profiles in tomato fruit constitutively expressing the grapevine stilbene synthase gene. Plant Biotechnol J 3(1):57–69. https://doi.org/10.1111/j.1467-7652.2004.00099.x

Grassi D, Desideri G, Ferri C (2010) Flavonoids: antioxidants against atherosclerosis. Nutrients 2(8):889–902. https://doi.org/10.3390/nu2080889

Gutierrez-Merino C, Lopez-Sanchez C, Lagoa R, K Samhan-Arias A, Bueno C, Garcia-Martinez V (2011) Neuroprotective actions of flavonoids. Curr Med Chem 18(8):1195–1212. https://doi.org/10.2174/092986711795029735

Hemavathi, Upadhyaya CP, Akula N, Young KE, Chun SC, Kim DH, Park SW (2010) Enhanced ascorbic acid accumulation in transgenic potato confers tolerance to various abiotic stresses. Biotechnol Lett 32(2):321–330. https://doi.org/10.1007/s10529-009-0140-0

Huang JC, Zhong YJ, Liu J, Sandmann G, Chen F (2013) Metabolic engineering of tomato for high-yield production of astaxanthin. Metab Eng 17:59–67. https://doi.org/10.1016/j.ymben.2013.02.005

Huang T, Joshi V, Jander G (2014) The catabolic enzyme methionine gamma-lyase limits methionine accumulation in potato tubers. Plant Biotechnol J 12(7):883–893. https://doi.org/10.1111/pbi.12191

Ikram NA, Ghaffar A, Khan AA, Nawaz F, Hussain A (2023) Optimizing iron seed priming for enhanced yield and biofortification of tomato. J Plant Nutr 46(12):2796–2810. https://doi.org/10.1080/01904167.2022.2160751

Institute of Medicine, Food and Nutrition Board (2010) Dietary reference intakes for calcium and vitamin D. National Academy Press

Institute of Medicine, Food and Nutrition Board (2000) Dietary reference intakes for vitamin C, vitamin E, selenium, and carotenoids. National Academy Press.

Institute of Medicine, Food and Nutrition Board (2001) Dietary reference intakes for vitamin A, vitamin K, arsenic, boron, chromium, copper, iodine, iron, manganese, molybdenum, nickel, silicon, vanadium, and zinc. National Academy Press.

Joint FAO/WHO (2004) expert consultation on human vitamin and mineral requirements. Vitamin and Mineral Requirements in Human Nutrition. Available from: https://apps.who.int/iris/bitstream/handle/10665/42716/9241546123.pdf

Kedishvili NY (2013) Enzymology of retinoic acid biosynthesis and degradation: thematic review series: fat-soluble vitamins: vitamin A. J Lipid Res 54(7):1744–60

Kromann P, Valverde F, Alvarado S, Vélez R, Pisuña J, Potosí B et al (2017) Can Andean potatoes be agronomically biofortified with iron and zinc fertilizers? Plant Soil 411(1–2):121–138. https://doi.org/10.1007/s11104-016-3065-0

Krzemińska J, Leszczyńska T, Kopeć A, and Piątkowska E, and Smoleń S, Kowalska I, Skoczeń-Słupska R, Słupski J, Koronowicz A (2025) Potato Tubers Biofortified with Iodoquinolines after Heat Treatment - Analysis of Quality Parameters, Bioavailability of Iodine and Selected Thyroid-Regulated Biochemical

Parameters in the Wistar Rats. Available at SSRN: https://ssrn.com/abstract=5359909 or https://doi.org/10.2139/ssrn.5359909

Ledwożyw-Smoleń I, Smoleń S, Rożek S, Sady W, Strzetelski P (2020) Iodine biofortification of potato (*Solanum tuberosum* L.) grown in field. Agronomy 10(12):1916. https://doi.org/10.3390/agronomy10121916

Li Q, Li Y, Li C, Yu X (2012) Enhanced ascorbic acid accumulation through overexpression of dehydroascorbate reductase confers tolerance to methyl viologen and salt stresses in tomato. Czech J Genet Plant Breed 48(2):74–86. https://doi.org/10.17221/100/2011-CJGPB

Li R, Li DW, Liu HP, Hong CL, Song MY, Dai ZX et al (2017) Enhancing iodine content and fruit quality of pepper (*Capsicum annuum* L.) through biofortification. Sci Hortic 214:165–173. https://doi.org/10.1016/j.scienta.2016.11.030

Li Y, Chu Z, Luo J, Zhou Y, Cai Y, Lu Y et al (2018) The C2H2 zinc-finger protein SlZF3 regulates AsA synthesis and salt tolerance by interacting with CSN5B. Plant Biotechnol J 16(6):1201–1213. https://doi.org/10.1111/pbi.12863

Li X, Ye J, Munir S, Yang T, Chen W, Liu G et al (2019) Biosynthetic gene pyramiding leads to ascorbate accumulation with enhanced oxidative stress tolerance in tomato. Int J Mol Sci 20(7):1558. https://doi.org/10.3390/ijms20071558

Li J, Scarano A, Gonzalez NM, D'Orso F, Yue Y, Nemeth K et al (2022) Biofortified tomatoes provide a new route to vitamin D sufficiency. Nat Plants 8(6):611–616. https://doi.org/10.1038/s41477-022-01154-6. (**PMID: 35606499**)

Lim MY, Pulla RK, Park JM, Harn CH, Jeong BR (2012) Overexpression of l-gulono-γ-lactone oxidase (GLOase) gene leads to ascorbate accumulation with enhanced abiotic stress tolerance in tomato. In Vitro Cell Dev Biol Plant 48(5):453–461. https://doi.org/10.1007/s11627-012-9461-0

Liu W, Feng Y, Yu S, Fan Z, Li X, Li J, Yin H (2021) The flavonoid biosynthesis network in plants. Int J Mole Sci 22(23):12824. https://doi.org/10.3390/ijms222312824

Lopez AB, Van Eck J, Conlin BJ, Paolillo DJ, O'Neill J, Li L (2008) Effect of the cauliflower or transgene on carotenoid accumulation and chromoplast formation in transgenic potato tubers. J Exp Bot 59(2):213–23

Lopez A, Cacoub P, Macdougall IC, Peyrin-Biroulet L (2016) Iron deficiency anaemia. Lancet 387(10021):907–916. https://doi.org/10.1016/S0140-6736(15)60865-0

Lu Y, Rijzaani H, Karcher D, Ruf S, Bock R (2013) Efficient metabolic pathway engineering in transgenic tobacco and tomato plastids with synthetic multigene operons. Proc Natl Acad Sci USA 110(8):E623–32. https://doi.org/10.1073/pnas.1216898110

Lukaszewicz M, Matysiak-Kata I, Skala J, Fecka I, Cisowski W, Szopa J (2004) Antioxidant capacity manipulation in transgenic potato tuber by changes in phenolic compounds content. J Agric Food Chem 52(6):1526–1533. https://doi.org/10.1021/jf034482k

Luo J, Butelli E, Hill L, Parr A, Niggeweg R, Bailey P, Weisshaar B, Martin C (2008) AtMYB12 regulates caffeoyl quinic acid and flavonol synthesis in tomato: expression in fruit results in very high levels of both types of polyphenol Summary. Plant J 56(2):316–326. https://doi.org/10.1111/j.1365-313X.2008.03597.x

Marriott BP, Birt DF, Stalling VA, Yates AA (eds) (2020) Present knowledge in nutrition: basic nutrition and metabolism. Academic Press

Maxfield L, Shukla S, Crane JS (2022) Zinc deficiency. StatPearls. Internet. In Stat. Pearls Publishing. Available from: http://www.ncbi.nlm.nih.gov/books/NBK493231/.

Mayer JE, Pfeiffer WH, Beyer P (2008) Biofortified crops to alleviate micronutrient malnutrition. Curr Opin Plant Biol 11(2):166–170. https://doi.org/10.1016/j.pbi.2008.01.007

Medrano-Macías J, Leija-Martínez P, González-Morales S, Juárez-Maldonado A, Benavides-Mendoza A (2016) Use of iodine to biofortify and promote growth and stress tolerance in crops. Front Plant Sci 7:1146. https://doi.org/10.3389/fpls.2016.01146

Moinuddin G, Jash S, Sarkar A, Dasgupta S (2017) Response of potato (*Solanum tuberosum* L.) to foliar application of macro and micronutrients in the red and lateritic zone of West Bengal. J Crop Weed 13(1):185–188

Muir SR, Collins GJ, Robinson S, Hughes S, Bovy A, Ric De Vos CH, van Tunen AJ, Verhoeyen ME (2001) Overexpression of petunia chalcone isomerase in tomato results in fruit containing increased levels of flavonols. Nat Biotechnol 19(5):470–474. https://doi.org/10.1038/88150

Mushtaq Z, Al-Ashkar I, Habib-Ur-Rahman M, Sabagh AE, Ilić P (2024) Biofortification of iron in potato through rhizobacteria and plant growth regulator. Potato Res 67(3):785–793. https://doi.org/10.1007/s11540-023-09667-z

Nestel P, Bouis HE, Meenakshi JV, Pfeiffer W (2006) Biofortification of staple food crops. J Nutr 136(4):1064–1067. https://doi.org/10.1093/jn/136.4.1064

Niggeweg R, Michael AJ, Martin C (2004) Engineering plants with increased levels of the antioxidant chlorogenic acid. Nat Biotechnol 22(6):746–754. https://doi.org/10.1038/nbt966

Noulas C, Tziouvalekas M, Karyotis T (2018) Zinc in soils, water and food crops. J Trace Elem Med Biol 49:252–260. https://doi.org/10.1016/j.jtemb.2018.02.009

Parra-Torrejón B, Cáceres A, Sánchez M, Sainz L, Guzmán M, Bermúdez-Perez FJ et al (2023) Multifunctional nanomaterials for biofortification and protection of tomato plants. Environ Sci Technol 57(40):14950–14960. https://doi.org/10.1021/acs.est.3c02559. (**PMID: 37753594**)

Pfeiffer WH, McClafferty B (2007) HarvestPlus: breeding crops for better nutrition. Crop Sci. 47(Suppl 3). https://doi.org/10.2135/cropsci2007.09.0020IPBS

Qaim M, Stein AJ, Meenakshi JV (2007) Economics of biofortification. Agric Econ 37(s1):119–133. https://doi.org/10.1111/j.1574-0862.2007.00239.x

Qin A, Shi Q, Yu X (2011) Ascorbic acid contents in transgenic potato plants overexpressing two dehydroascorbate reductase genes. Mol Biol Rep 38(3):1557–1566. https://doi.org/10.1007/s11033-010-0264-2. (**PMID: 20857222**)

Rabbi R, Chowdhury M, Saha B (2022) Agronomic approaches to biofortify iron in tomato. J Bangladesh Agril Univ. https://doi.org/10.5455/JBAU.120912s

Rahim FP, Rocio CG, Adalberto BM, Lidia Rosaura SC, Maginot NH (2020) Agronomic biofortification with selenium in tomato crops (Solanum lycopersicon L. Mill). Agriculture 10(10):486. https://doi.org/10.3390/agriculture10100486

RaiesiArdali T, Ma'mani L, Chorom M, Motamedi E, FathiGharebaba M (2024) A biocompatible NPK+Fe+Zn slow release fertilizer: synthesis and its evaluation in tomato plant growth improvement. Sci Rep. https://doi.org/10.1038/s41598-024-55152-z. (**PMID: 38409209**)

Raiesi-Ardali T, Ma'mani L, Chorom M, Moezzi A (2022) Improved iron use efficiency in tomato using organically coated iron oxide nanoparticles as efficient bioavailable Fe sources. Chem Biol Technol Agric. https://doi.org/10.1186/s40538-022-00318-y

Ren W, Qiao Z, Wang H, Zhu L, Zhang L (2003) Flavonoids: promising anticancer agents. Med Res Rev 23(4):519–534. https://doi.org/10.1002/med.10033

Roberts RJ, Knight ME (1987) Pharmacology of Vitamin E in the newborn. Clin Perinatol 14(4):843–855. https://doi.org/10.1016/S0095-5108(18)30737-1

Römer S, Fraser PD, Kiano JW, Shipton CA, Misawa N, Schuch W, Bramley PM (2000) Elevation of the provitamin A content of transgenic tomato plants. Nat Biotechnol 18(6):666–9

Römer S, Lübeck J, Kauder F, Steiger S, Adomat C, Sandmann G (2002) Genetic Engineering of a Zeaxanthin-rich Potato by

Antisense Inactivation and Co-suppression of Carotenoid Epoxidation. Metab Eng 4(4):263–272. https://doi.org/10.1006/mben.2002.0234

Rosati C, Aquilani R, Dharmapuri S, Pallara P, Marusic C, Tavazza R, Bouvier F, Camara B, Giuliano G (2000) Metabolic engineering of beta-carotene and lycopene content in tomato fruit. Plant J 24(3):413–20

Ross A, Vitamin A (2014). In: Ross AC, Caballero B, Cousins RJ, Tucker KL, Ziegler TR (eds) Modern nutrition in health and disease. Lippincott, Williams & Wilkins, pp 260–277

Ryu M-S, Aydemir TB (2020) Zinc. In: Marriott BP, Birt DF, Stallings VA, Yates AA, editors. Present knowledge in nutrition. Wiley-Blackwell. pp. 393-408. https://doi.org/10.1016/B978-0-323-66162-1.00023-8

Sabatino L, La Bella S, Ntatsi G, Iapichino G, D'Anna F, De Pasquale C, Consentino BB, Rouphael Y (2021) Selenium biofortification and grafting modulate plant performance and functional features of cherry tomato grown in a soilless system. Sci Hortic 285:110095. https://doi.org/10.1016/j.scienta.2021.110095

Saltzman A, Birol E, Bouis HE, Boy E, De Moura FF, Islam Y et al (2013) Biofortification: progress toward a more nourishing future. Glob Food Secur 2(1):9–17. https://doi.org/10.1016/j.gfs.2012.12.003

Shiriaev A, Pezzarossa B, Rosellini I, Malorgio F, Lampis S, Ippolito A et al (2022) Efficacy and comparison of different strategies for selenium biofortification of tomatoes. Horticulturae 8(9):800. https://doi.org/10.3390/horticulturae8090800

Skalny AV, Aschner M, Tinkov AA (2021) Zinc Adv Food Nutr Res 96:251–310. https://doi.org/10.1016/bs.afnr.2021.01.003. (**PMID: 34112355**)

Smoleń S, Wierzbińska J, Sady W, Kołton A, Wiszniewska A, Liszka-Skoczylas M (2015) Iodine biofortification with additional application of salicylic acid affects yield and selected parameters of chemical composition of tomato fruits (*Solanum lycopersicum* L.). Sci Hortic 188:89–96

Somma A, Palmitessa OD, Conversa G, Serio F, Santamaria P (2024) Localized foliar application of iodine on tomato: an effective approach for targeted agronomic biofortification. Sci Hortic 1(327):112807

Song XY, Zhu WJ, Tang RM, Cai JH, Chen M, Yang Q (2016) Overexpression of StLCYb increases β-carotene accumulation in potato tubers. Plant Biotechnol Rep 10(2):95–104

Thibeault DW (2000) The precarious antioxidant defenses of the preterm infant. Am J Perinatol 17(4):167–182. https://doi.org/10.1055/s-2000-9422

Van Eck JO, Conlin BR, Garvin DF, Mason H, Navarre DA, Brown CR (2007) Enhancing beta-carotene content in potato by RNAi-mediated silencing of the beta-carotene hydroxylase gene. Am J Potato Res 84(4):331–42

Vergara Carmona VM, Cecílio Filho AB, Almeida HJ, Gratão PL (2019) Fortification and bioavailability of zinc in potato. J Sci Food Agric 99(7):3525–3529. https://doi.org/10.1002/jsfa.9572

Wallace A (1988) Acid and acid–iron fertilizers for iron-deficiency control in plants. J Plant Nutr 11(6-11):1311–1319. https://doi.org/10.1080/01904168809363888

Wevar Oller ALW, Agostini E, Milrad SR, Medina MI (2009) In situ and de novo biosynthesis of vitamin C in wild type and transgenic tomato hairy roots: a precursor feeding study. Plant Sci 177(1):28–34. https://doi.org/10.1016/j.plantsci.2009.03.017

White PJ, Thompson JA, Wright G, Rasmussen SK (2017) Biofortifying Scottish potatoes with zinc. Plant Soil 411(1–2):151–165. https://doi.org/10.1007/s11104-016-2903-4

WHO (2001) Iron deficiency anaemia: assessment. Prev Control. Available from: https://www.who.int/publications/m/item/iron-children-6to23--archived-iron-deficiency-anaemia-assessment-prevention-and-control.

WHO (2009) Global prevalence of vitamin A deficiency in populations at risk 1995-2005. WHO global database on vitamin A deficiency. World Health Organization. Available from: http://whqlibdoc.who.int/publications/2009/9789241598019_eng.pdf

WHO (2013) Vitamin and Mineral Nutrition Information System (VMNIS) micronutrients database. Available from: http://www.who.int/vmnis/database/en/.

Wirth JP, Petry N, Tanumihardjo SA, Rogers LM, McLean E, Greig A, Garrett GS, Klemm RD, Rohner F (2017) Vitamin A supplementation programs and country-level evidence of vitamin A deficiency. Nutrients 9(3):190

Wurbs D, Ruf S, Bock R (2007) Contained metabolic engineering in tomatoes by expression of carotenoid biosynthesis genes from the plastid genome. Plant J 49(2):276–88

Ye J, Liu G, Chen W, Zhang F, Li H, Ye Z, Zhang Y (2019) Knockdown of SlNL33 accumulates ascorbate enhances disease and oxidative stress tolerance in tomato (Solanum lycopersicum). Plant Growth Regul 89(1):49–58. https://doi.org/10.1007/s10725-019-00512-3

Zeh M, Casazza AP, Kreft O, Roessner U, Bieberich K, Willmitzer L et al (2001) Antisense inhibition of threonine synthase leads to high methionine content in transgenic potato plants. Plant Physiol 127(3):792–802. https://doi.org/10.1104/pp.010438

Zha M, Li X, Li R, Huang J, Fan J, Zhang J et al (2022) Overexpression of nicotianamine synthase (AtNAS1) increases iron accumulation in the tuber of potato. Plants 11(20):2741. https://doi.org/10.3390/plants11202741

Zhang C, Liu J, Zhang Y, Cai X, Gong P, Zhang J, Wang T, Li H, Ye Z (2011) Overexpression of SlGMEs leads to ascorbate accumulation with enhanced oxidative stress cold and salt tolerance in tomato. Plant Cell Rep 30(3):389–398. https://doi.org/10.1007/s00299-010-0939-0

Zhang R, Zhang W, Kang Y, Shi M, Yang X, Li H et al (2022) Application of different foliar iron fertilizers for improving the photosynthesis and tuber quality of potato (*Solanum tuberosum* L.) and enhancing iron biofortification. Chem Biol Technol Agric. https://doi.org/10.1186/s40538-022-00346-8

Zimmermann MB, Jooste PL, Pandav CS (2008) Iodine-deficiency disorders. Lancet 372(9645):1251–1262. https://doi.org/10.1016/S0140-6736(08)61005-3

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.